2. It consists of two monomeric units plus J chain and secretory component (Figure 13.29).
3. The dominant subclass of sIgA is sIgA2 which is unique for its absence of a covalent bond between the light and heavy chains. In this subclass, light chains are linked by disulphide bonds.
4. It has a half life of 5-6 days. It is responsible for local immunity.
5. The sIgA molecules protect mucosal surfaces by reacting with the surface of potential pathogens and interfering with their adherence and colonization. It also plays a role in the alternative complement pathway.

iii) IgM

1. IgM accounts for about 5-10% of the serum immunoglobulin pool.
2. It has a pentameric structure consisting of five monomeric units linked by J chain and disulphide bonds at the Fc fragment (Figure 13.23).
3. It is the predominant antibody in the primary immune response to most antigens and the predominant antibody produced by the fetus.
4. It is the first immunoglobulin made during B cell maturation and individual IgM monomers are expressed on B cells, serving as the antibody component of the B cell receptor (BCR).
5. IgM tends to remain in the bloodstream, where it agglutinates (clumps) bacteria, activates complement by the classical pathway and enhances the ingestion of pathogens by phagocytic cells.

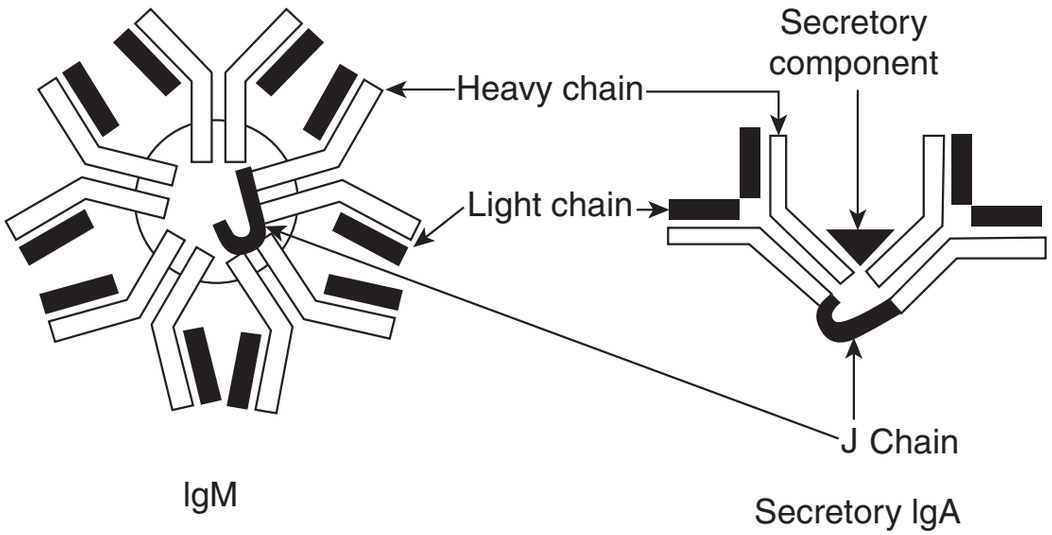

Figure 13.27: Structural models of IgM and secretory IgA. IgM has a pentameric structure linked by J chain at the Fc fragment. Secretory IgA has a dimeric structure plus J chain plus secretory component and is shown in the dominant IgA2 subclass, which is unique for its absence of a convalent bond between the light and heavy chains. Light chains are linked by disulfide bonds.

TN_GOVT_XI_Micro_Biology_CH13.indd 242 16-02-2019 10.36.11 AM